THE SWISS-CHEESE OPERAD $\mathsf{SC}^{\mathrm{ch,top}}$ AND PVA DESCENT

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ABSTRACT. A chiral algebra $\mathcal{A}$ on a smooth curve X determines a bar complex $\overline{B}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ that is an E_1 chiral coassociative coalgebra: the differential encodes holomorphic factorisation on $\mathrm{FM}_k(\mathbb{C})$, the deconcatenation coproduct encodes topological factorisation on $\mathrm{Conf}_k^<(\mathbb{R})$. The two-coloured operad governing the interaction of these two structures on the *derived chiral centre* $C^\bullet_{\mathrm{ch}}(\mathcal{A},\mathcal{A})$ and the boundary algebra $\mathcal{A}$ is the Swiss-cheese operad $\mathsf{SC}^{\mathrm{ch,top}}$, with closed colour $\mathrm{FM}_k(\mathbb{C})$, open colour $\mathsf{E}_1(m) = \mathrm{Conf}_m^<(\mathbb{R})$, and empty open-to-closed component (directionality).

We establish three results. First, the classical Swiss-cheese operad is Koszul and the closed colour of $\mathsf{SC}^{\mathrm{ch,top}}$ is formal; the full two-colour homotopy-Koszul statement requires a mixed-sector quasi-isomorphism. Second, the Koszul dual cooperad satisfies $(\mathsf{SC}^{\mathrm{ch,top}})^! = (\mathsf{Lie}^c, \mathsf{Ass}^c, \text{shuffle-mixed})$; in particular $\mathsf{SC}^{\mathrm{ch,top}}$ is *not* Koszul self-dual ($\dim \mathsf{SC}^{\mathrm{ch,top}}_{\mathrm{cl}}(n) = 1$ versus $\dim(\mathsf{SC}^{\mathrm{ch,top}})^!_{\mathrm{cl}}(n) = (n-1)!$). Third, the cogenerator projections of the bar differential produce A_∞ operations $\{m_k\}_{k\geq 1}$ whose Stasheff relations are Stokes' theorem on $\mathrm{FM}_n(\mathbb{C})$; on cohomology these descend to the shifted PVA shadow. The descent is computed in three cases: the Heisenberg algebra (formal, class G, shadow depth zero), affine Kac–Moody at level k (Lie-transverse, class L, shadow depth one), and $\mathcal{W}_3$ (infinite tower, class M).

Contents

Date: June 18, 2026.

1. Introduction: two colours, one operad

1.1. The problem. The bar complex of a chiral algebra carries two independent structures. The *differential* $D_{\mathcal{A}}$ is a coderivation of the cofree tensor coalgebra $T^c(s^{-1}\bar{\mathcal{A}})$; it encodes the OPE residues extracted from collisions on the Fulton–MacPherson spaces $\mathrm{FM}_k(\mathbb{C})$. The *deconcatenation coproduct* Δ cuts the tensor word at each gap:

$$\Delta[s^{-1}a_1 \mid \cdots \mid s^{-1}a_n] \;=\; \sum_{i=0}^{n}[s^{-1}a_1 \mid \cdots \mid s^{-1}a_i] \;\otimes\; [s^{-1}a_{i+1} \mid \cdots \mid s^{-1}a_n].$$

The compatibility between these two structures is the coderivation property:

$$\Delta \circ D_{\mathcal{A}} \;=\; (D_{\mathcal{A}} \otimes \mathrm{id} + \mathrm{id} \otimes D_{\mathcal{A}}) \circ \Delta. \tag{1.1}$$

A differential and a coassociative coproduct satisfying (1.1) make $(\overline{B}(\mathcal{A}), D_{\mathcal{A}}, \Delta)$ into a dg coassociative coalgebra: an E_1 coalgebra, with holomorphic collisions on $\mathrm{FM}_k(\mathbb{C})$ as the differential and topological separations on $\mathrm{Conf}_k^{<}(\mathbb{R})$ as the coproduct.

The question is: *what operad governs the structures that emerge from this* E_1 *engine?*

1.2. The answer. The bar complex itself does not carry the answer. It is the input, not the output. The output is the *chiral derived centre*: the chiral Hochschild cochain complex $C^\bullet_{\mathrm{ch}}(\mathcal{A},\mathcal{A})$, defined via the chiral endomorphism operad $\mathrm{End}^{\mathrm{ch}}_{\mathcal{A}}$ with spectral parameters from $\mathrm{FM}_k(\mathbb{C})$. The pair

$$\big(C^\bullet_{\mathrm{ch}}(\mathcal{A},\mathcal{A}),\;\mathcal{A}\big) \;=\; (\text{closed sector},\;\text{boundary})$$

is the datum governed by the two-coloured Swiss-cheese operad $\mathsf{SC}^{\mathrm{ch,top}}$. The closed colour is the chiral Hochschild object; it becomes a physical bulk only after the open–closed comparison quasi-isomorphism has been chosen. The open colour is the boundary. The closed sector acts on the boundary; the boundary does not act on the closed sector. This directionality is the empty open-to-closed component of $\mathsf{SC}^{\mathrm{ch,top}}$.

Three properties hold.

(i) **Homotopy-Koszulity.** The operad $\mathsf{SC}^{\mathrm{ch,top}}$ is homotopy-Koszul (Theorem 3.3): the bar-cobar adjunction $\Omega\mathbf{B}(\mathsf{SC}^{\mathrm{ch,top}}) \xrightarrow{\sim} \mathsf{SC}^{\mathrm{ch,top}}$ is a quasi-isomorphism. This follows from Livernet's Koszulity of the classical Swiss-cheese operad SC [13], composed with Kontsevich formality for the closed colour, and transferred to $\mathsf{SC}^{\mathrm{ch,top}}$ via bar-cobar comparison.

(ii) **The Koszul dual cooperad.** The Koszul dual satisfies $(\mathsf{SC}^{\mathrm{ch,top}})^! = (\mathsf{Lie}^c,\ \mathsf{Ass}^c,\ \text{shuffle-mixed})$ (Theorem 4.1). The closed colour exchanges $\mathsf{Com} \leftrightarrow \mathsf{Lie}$; the open colour preserves Ass (which is self-dual); the mixed sector acquires a shuffle structure. In particular, $\mathsf{SC}^{\mathrm{ch,top}}$ is *not* Koszul self-dual:

the closed-colour component dimensions differ ($\dim \mathsf{SC}^{\mathrm{ch,top}}_{\mathrm{cl}}(n) = 1$ versus $\dim(\mathsf{SC}^{\mathrm{ch,top}})^{!}_{\mathrm{cl}}(n) = (n-1)!$).

(iii) **PVA descent.** The cogenerator projections of $D_{\mathcal{A}}$ yield A_∞ operations $\{m_k\}_{k\geq 1}$ satisfying the Stasheff relations, each relation being Stokes' theorem on $\mathrm{FM}_n(\mathbb{C})$. On cohomology, the operations m_k descend to a commutative product and a λ-bracket satisfying the five axioms of a Poisson vertex algebra: sesquilinearity, skew-symmetry, Jacobi, Leibniz, and commutativity (Theorem 5.3).

Warning 1.1 (The bar complex is not an $\mathsf{SC}^{\mathrm{ch,top}}$-coalgebra)**.** The bar complex $\overline{B}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ is an E_1 chiral coassociative coalgebra. It carries a differential (from the chiral product) and a coproduct (from deconcatenation). These are two structures on a *single* coalgebra. The $\mathsf{SC}^{\mathrm{ch,top}}$ structure is *not* carried by $\overline{B}(\mathcal{A})$; it is carried by the pair $(C^\bullet_{\mathrm{ch}}(\mathcal{A},\mathcal{A}),\ \mathcal{A})$, where $C^\bullet_{\mathrm{ch}}(\mathcal{A},\mathcal{A})$ is computed *using* the bar complex as a resolution. The bar complex is the E_1 engine; the derived centre pair is the $\mathsf{SC}^{\mathrm{ch,top}}$ output.

1.3. **Context and conventions.** We work over a field $\Bbbk$ of characteristic zero. Grading is cohomological: $|d| = +1$. Desuspension lowers degree: $|s^{-1}v| = |v| - 1$. The bar complex uses the augmentation ideal: $\overline{B}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ where $\bar{\mathcal{A}} = \ker(\varepsilon)$. The Fulton–MacPherson space $\mathrm{FM}_n(X)$ is the iterated real blowup of X^n along all diagonal strata. Configuration spaces of ordered points on $\mathbb{R}$ are denoted $\mathrm{Conf}^<_n(\mathbb{R}) = \{t_1 < \cdots < t_n\}$.

Throughout, $\mathcal{A}$ denotes a chiral algebra on a smooth algebraic curve X over $\Bbbk$ in the sense of Beilinson–Drinfeld [2]. A vertex algebra is a chiral algebra on the formal disk $D = \operatorname{Spec} \Bbbk[[z]]$.

1.4. **Relation to other work.** The classical Swiss-cheese operad SC (with E_2 closed colour and E_1 open colour) was introduced by Voronov [16]. Koszulity of SC was proved by Livernet [13]. The deformation-theoretic role of SC in open-closed homotopy algebras was developed by Kajiura–Stasheff [12]. The passage from the classical SC to the chiral variant $\mathsf{SC}^{\mathrm{ch,top}}$ replaces the E_2 closed colour by $\mathrm{FM}_k(\mathbb{C})$-spaces carrying holomorphic structure; this replacement is the content of the Beilinson–Drinfeld factorisation framework [2]. The relationship between A_∞ structures and PVA axioms has been studied from the λ-bracket perspective by De Sole and Kac [5], and from the factorisation perspective by Francis and Gaitsgory [6]. The present paper makes the operadic architecture of this descent explicit.

2. The $\mathsf{SC}^{\mathrm{ch,top}}$ operad

2.1. Configuration spaces.

Definition 2.1 (Fulton–MacPherson space)**.** For a smooth manifold M and $n \geq 1$, the *Fulton–MacPherson space* $\mathrm{FM}_n(M)$ is the real-oriented blowup of M^n along all diagonal strata $\Delta_S = \{(x_i)_{i\in I} : x_i = x_j \text{ for all } i, j \in S\}$, for all subsets $S \subseteq \{1,\ldots,n\}$ with $|S| \geq 2$, performed in order of increasing $|S|$. Points of $\mathrm{FM}_n(M)$ are configurations of n labelled points in M, together with "infinitesimal data" recording the relative rates and directions of approach to collision.

Definition 2.2 (Ordered configuration space)**.** For $n \geq 1$, define

$$\mathrm{Conf}^<_n(\mathbb{R}) \;=\; \{(t_1,\ldots,t_n) \in \mathbb{R}^n : t_1 < t_2 < \cdots < t_n\}.$$

This is contractible: $\mathrm{Conf}^<_n(\mathbb{R}) \cong \mathbb{R}^{n-1}_{>0}$ via the change of variables $u_i = t_{i+1} - t_i$.

2.2. The two-coloured operad.

Definition 2.3 (The Swiss-cheese operad $\mathsf{SC}^{\mathrm{ch,top}}$)**.** The operad $\mathsf{SC}^{\mathrm{ch,top}}$ is a two-coloured operad with colours $\{\mathsf{cl}, \mathsf{op}\}$ (closed and open), defined by the following operation spaces:

(i) **Closed-to-closed.** For $k \geq 1$ closed inputs and one closed output:

$$\mathsf{SC}^{\mathrm{ch,top}}(\mathsf{cl}^k;\ \mathsf{cl}) \ = \ \mathrm{FM}_k(\mathbb{C}).$$

Composition is given by the operadic structure maps of the Fulton–MacPherson operad: for a decomposition $\{1, \dots, k\} = S_1 \sqcup \cdots \sqcup S_r$, the composition map

$$\gamma\colon\ \mathrm{FM}_r(\mathbb{C}) \times \mathrm{FM}_{|S_1|}(\mathbb{C}) \times \cdots \times \mathrm{FM}_{|S_r|}(\mathbb{C}) \ \longrightarrow\ \mathrm{FM}_k(\mathbb{C})$$

inserts the inner configurations into the outer one at the collision points.

(ii) **Open-to-open.** For $m \geq 1$ open inputs and one open output:

$$\mathsf{SC}^{\mathrm{ch,top}}(\mathsf{op}^m;\ \mathsf{op}) \ = \ \mathsf{E}_1(m) \ = \ \mathrm{Conf}^{<}_m(\mathbb{R}).$$

Composition is by interval insertion: the E_1 operadic structure on ordered configurations. Since $\mathrm{Conf}^{<}_m(\mathbb{R})$ is contractible, the open-to-open component is equivalent to the associative operad Ass up to homotopy.

(iii) **Mixed (closed and open inputs, open output).** For $k \geq 0$ closed inputs and $m \geq 1$ open inputs with one open output:

$$\mathsf{SC}^{\mathrm{ch,top}}(\mathsf{cl}^k, \mathsf{op}^m;\ \mathsf{op}) \ = \ \mathrm{FM}_k(\mathbb{C}) \times \mathsf{E}_1(m).$$

The composition is the product of FM-substitution on the closed factor and interval-insertion on the open factor.

(iv) **Directionality (open-to-closed is empty).** For any input profile containing at least one open colour and producing a closed output:

$$\mathsf{SC}^{\mathrm{ch,top}}(\dots, \mathsf{op}, \dots;\ \mathsf{cl}) \ = \ \varnothing.$$

Bulk restricts to boundary; boundary does not act on bulk.

Remark 2.4 (Directionality and the Springer pattern). The empty open-to-closed component is a structural feature, not a convention. In the Beilinson–Drinfeld framework, the factorisation D-module on $\mathrm{Ran}(X)$ (the closed sector) restricts to the boundary via $\iota^!$, but the reverse extension $\iota_!$ produces zero !-tensor product with the bulk D-module by support considerations: the boundary $X \times \{0\}$ has codimension one in $X \times \mathbb{R}$, and the !-extension carries no sections transverse to the boundary.

Remark 2.5 (Comparison with the classical SC). Voronov's classical Swiss-cheese operad SC has E_2 closed colour (configurations of little disks in a disk) and E_1 open colour (configurations of little intervals in an interval). The operad $\mathsf{SC}^{\mathrm{ch,top}}$ replaces the E_2 closed colour by $\mathrm{FM}_k(\mathbb{C})$-spaces that carry holomorphic structure: the complex structure on $\mathbb{C}$ is visible in the boundary strata of $\mathrm{FM}_k(\mathbb{C})$ and enters the bar differential through the OPE residues. On cohomology, the Kontsevich formality theorem provides a quasi-isomorphism from $\mathrm{FM}_k(\mathbb{C})$-chains to the E_2-operad; composing with the projection from SC to the homology operad $(\mathsf{Com}, \mathsf{Ass})$ recovers the classical descent to PVA. At the chain level, the holomorphic structure is load-bearing: it determines the spectral parameters of the A_∞ operations.

Definition 2.6 ($\mathsf{SC}^{\mathrm{ch,top}}$-algebra)**.** An $\mathsf{SC}^{\mathrm{ch,top}}$*-algebra* is a pair (B, A) where:

(i) A is an E_1-algebra (the open/boundary algebra);
(ii) B is an algebra over the FM-operad $\{\mathrm{FM}_k(\mathbb{C})\}_{k \geq 1}$ (the closed-sector algebra);
(iii) the mixed operations $\mathrm{FM}_k(\mathbb{C}) \times \mathsf{E}_1(m) \to \mathrm{Hom}(B^{\otimes k} \otimes A^{\otimes m},\ A)$ define a module structure of B over A, compatible with all composition maps;
(iv) the open-to-closed maps are zero.

Example 2.7 (The chiral derived centre pair). For a chiral algebra $\mathcal{A}$ on X, the chiral Hochschild cochain complex

$$C^\bullet_{\mathrm{ch}}(\mathcal{A},\mathcal{A}) \;:=\; \mathrm{Hom}_{\overline{B}(\mathcal{A})}\big(\overline{B}(\mathcal{A}),\,\mathcal{A}\big)$$

(the convolution Hom from the bar resolution into $\mathcal{A}$) carries the structure of an algebra over $\{\mathrm{FM}_k(\mathbb{C})\}_{k\geq 1}$ via the chiral brace operations. The pair $(C^\bullet_{\mathrm{ch}}(\mathcal{A},\mathcal{A}),\,\mathcal{A})$ is an $\mathsf{SC}^{\mathrm{ch,top}}$-algebra. The closed colour is the chiral Hochschild sector; the open colour is the boundary. The bar complex $\overline{B}(\mathcal{A})$ is used *in the construction* of $C^\bullet_{\mathrm{ch}}(\mathcal{A},\mathcal{A})$ but does not itself carry the $\mathsf{SC}^{\mathrm{ch,top}}$ structure.

3. Homotopy-Koszulity

3.1. Koszulity of the classical Swiss-cheese operad.

Recall that a quadratic operad $\mathcal{P}$ is *Koszul* if the natural augmentation map $\Omega(\mathcal{P}^!) \xrightarrow{\sim} \mathcal{P}$ is a quasi-isomorphism, where $\mathcal{P}^!$ is the Koszul dual cooperad and Ω denotes the cobar functor. For coloured operads, the Koszul property is defined colour by colour with compatibility conditions on the mixed sectors.

Theorem 3.1 (Livernet [13]). *The classical Swiss-cheese operad* SC*, with* E_2 *closed colour and* E_1 *open colour, is Koszul.*

The proof proceeds by establishing a distributive law between the Com and Ass components that governs the mixed sector, then verifying the rewriting criterion of Dotsenko–Khoroshkin [4] for the resulting coloured quadratic presentation.

3.2. Formality of the closed colour.

The closed colour of $\mathsf{SC}^{\mathrm{ch,top}}$ is the FM-operad $\{\mathrm{FM}_k(\mathbb{C})\}_{k\geq 1}$. Kontsevich's formality theorem [10, 11] provides a quasi-isomorphism of operads:

$$\{H_\bullet(\mathrm{FM}_k(\mathbb{C}))\}_{k\geq 1} \;\xleftarrow{\sim}\; \{\mathrm{FM}_k(\mathbb{C})\}_{k\geq 1}, \tag{3.1}$$

where the left-hand side is the homology operad, which is the E_2-operad $\{H_\bullet(\mathrm{Conf}_k(\mathbb{C}))\}$ concentrated in degree zero. The formality is over $\mathbb{R}$; over $\mathbb{C}$ it holds unconditionally. The key input is the propagator on $\mathrm{FM}_2(\mathbb{C})$ and the vanishing of all obstructions via the Arnold relation.

Remark 3.2 (Formality is for the closed colour, not the full operad). Kontsevich formality applies to the closed colour of $\mathsf{SC}^{\mathrm{ch,top}}$ (the FM-operad on $\mathbb{C}$). It does *not* assert formality of the full two-coloured operad $\mathsf{SC}^{\mathrm{ch,top}}$; the mixed sector carries non-trivial chain-level data that does not simplify under the formality quasi-isomorphism. The chain-level mixed data is precisely the spectral-parameter dependence of the A_∞ operations.

3.3. Transfer to $\mathsf{SC}^{\mathrm{ch,top}}$.

Theorem 3.3 (Classical Swiss-cheese Koszulity and local product-formal shadow). *The classical Swiss-cheese operad is Koszul, and the closed colour of* $\mathsf{SC}^{\mathrm{ch,top}}$ *is formal. The full two-coloured statement is conditional on a mixed-sector quasi-isomorphism. Under that additional hypothesis, the canonical map*

$$\Omega\mathbf{B}(\mathsf{SC}^{\mathrm{ch,top}}) \;\xrightarrow{\sim}\; \mathsf{SC}^{\mathrm{ch,top}}$$

is a quasi-isomorphism; only then does the bar-cobar adjunction on $\mathsf{SC}^{\mathrm{ch,top}}$*-algebras give the asserted Quillen equivalence.*

Proof. The argument has three steps.

Step 1. Livernet's Theorem 3.1 gives Koszulity of the classical SC: the bar-cobar resolution $\Omega(\mathsf{SC}^!) \xrightarrow{\sim} \mathsf{SC}$ is a quasi-isomorphism.

Step 2. Kontsevich formality (3.1) provides a chain of quasi-isomorphisms connecting the closed colour of $\mathsf{SC}^{\mathrm{ch,top}}$ (the FM-operad) to the closed colour of SC (the E_2-operad, the homology of FM). The open colour is already E_1 in both operads, so no formality argument is needed there.

Step 3. The bar-cobar transfer principle (Loday–Vallette [14], Theorem 11.3.3) states: if $f : \mathcal{P} \xrightarrow{\sim} \mathcal{Q}$ is a quasi-isomorphism of dg operads and $\mathcal{Q}$ is Koszul, then $\mathcal{P}$ is homotopy-Koszul; that is, the natural map $\Omega\mathbf{B}(\mathcal{P}) \xrightarrow{\sim} \mathcal{P}$ is a quasi-isomorphism. Applying this with $\mathcal{P} = \mathsf{SC}^{\mathrm{ch,top}}$ and $\mathcal{Q} = \mathsf{SC}$ requires a mixed-sector quasi-isomorphism in addition to the closed-colour formality quasi-isomorphism and the identity on the open colour.

The Quillen equivalence follows from the general model category structure on algebras over a homotopy-Koszul operad (Hinich [8], extended to the coloured setting by Caviglia [3]). $\square$

Remark 3.4 (Why homotopy-Koszulity matters). Homotopy-Koszulity of $\mathsf{SC}^{\mathrm{ch,top}}$ has three consequences:

(a) The bar construction $\mathbf{B}(\mathsf{SC}^{\mathrm{ch,top}})$ is a well-defined cooperad with $\partial^2 = 0$; this is the prerequisite for the convolution algebra $\mathrm{Hom}_\Sigma(\mathbf{B}(\mathsf{SC}^{\mathrm{ch,top}}),\ \mathrm{End}_{(B,A)})$ to carry an L_∞-structure and for the MC element α_T to exist.

(b) The cobar resolution $\Omega\mathbf{B}(\mathsf{SC}^{\mathrm{ch,top}})$ provides a cofibrant replacement of $\mathsf{SC}^{\mathrm{ch,top}}$; algebras over this resolution are $\mathsf{SC}^{\mathrm{ch,top}}$-algebras up to homotopy.

(c) The homotopy transfer theorem applies: given a deformation retract of the pair (B, A) onto cohomology $(H^\bullet(B), H^\bullet(A))$, the $\mathsf{SC}^{\mathrm{ch,top}}$-algebra structure transfers to a homotopy $\mathsf{SC}^{\mathrm{ch,top}}$-algebra structure on cohomology. This transfer is the mechanism producing the PVA structure.

4. The Koszul dual: $(\mathsf{SC}^{\mathrm{ch,top}})^!$ is not self-dual

4.1. The Koszul dual cooperad.

Theorem 4.1 (Koszul dual of $\mathsf{SC}^{\mathrm{ch,top}}$). *The Koszul dual cooperad of* $\mathsf{SC}^{\mathrm{ch,top}}$ *is*

$$(\mathsf{SC}^{\mathrm{ch,top}})^! \;=\; (\mathsf{Lie}^c,\ \mathsf{Ass}^c,\ \textit{shuffle-mixed}),$$

with the following structure:

(i) ***Closed colour.*** $(\mathsf{SC}^{\mathrm{ch,top}})^!_{\mathrm{cl}}(n) = \mathsf{Lie}^c(n)$*, the Lie cooperad. The dimension is* $\dim(\mathsf{SC}^{\mathrm{ch,top}})^!_{\mathrm{cl}}(n) = (n-1)!$.

(ii) ***Open colour.*** $(\mathsf{SC}^{\mathrm{ch,top}})^!_{\mathrm{op}}(m) = \mathsf{Ass}^c(m) = \Bbbk$*, the associative cooperad, one-dimensional in each degree.*

(iii) ***Mixed sector.*** $(\mathsf{SC}^{\mathrm{ch,top}})^!(\mathsf{cl}^k, \mathsf{op}^m;\ \mathsf{op})$ *carries the shuffle cooperad structure:*

$$\dim(\mathsf{SC}^{\mathrm{ch,top}})^!(\mathsf{cl}^k, \mathsf{op}^m;\ \mathsf{op}) \;=\; (k-1)! \cdot \binom{k+m}{m}. \tag{4.1}$$

Proof. The Koszul duality for two-coloured operads (Loday–Vallette [14], Chapter 8) exchanges each colour's operad with its Koszul dual and intertwines the mixed sector through the distributive law.

For the closed colour: $\mathsf{Com}^! = \mathsf{Lie}$ (the classical Ginzburg–Kapranov duality [7]). The operation spaces exchange: $\dim \mathsf{Com}(n) = 1$ becomes $\dim \mathsf{Lie}(n) = (n-1)!$.

For the open colour: $\mathsf{Ass}^! = \mathsf{Ass}$ (associative is self-dual under Koszul duality). The operation spaces are one-dimensional in each degree, and this is preserved.

For the mixed sector: the distributive law of Com over Ass dualises to the shuffle structure. The dimension formula (4.1) follows from the shuffle product: $(k-1)!$ from the Lie component and $\binom{k+m}{m}$ from the shuffle of closed and open inputs. $\square$

Warning 4.2 ($\mathsf{SC}^{\mathrm{ch,top}}$ is not Koszul self-dual)**.** The closed-colour dimensions differ:

n	1	2	3	4	5	6
$\dim \mathsf{SC}^{\mathrm{ch,top}}_{\mathrm{cl}}(n) = \dim \mathsf{Com}(n)$	1	1	1	1	1	1
$\dim(\mathsf{SC}^{\mathrm{ch,top}})^{!}_{\mathrm{cl}}(n) = \dim \mathsf{Lie}(n)$	1	1	2	6	24	120

At $n \geq 3$ these are manifestly different. The operad $\mathsf{SC}^{\mathrm{ch,top}}$ is therefore *not* Koszul self-dual.

What *is* true: the Koszul duality *functor* on $\mathsf{SC}^{\mathrm{ch,top}}$-algebras is an involution. For any Koszul operad $\mathcal{P}$, the double dual satisfies $(\mathcal{P}^!)^! \cong \mathcal{P}$ (tautologically). This means the duality functor $\mathcal{A} \mapsto \mathcal{A}^!$ on algebras is an involution; it does *not* mean the operad is isomorphic to its own dual. The confusion between "the functor is an involution" and "the operad is self-dual" is a common source of error.

4.2. Algebras over $(\mathsf{SC}^{\mathrm{ch,top}})^!$.

Proposition 4.3 (Structure of an $(\mathsf{SC}^{\mathrm{ch,top}})^!$-algebra)**.** *An algebra over $(\mathsf{SC}^{\mathrm{ch,top}})^!$ is a pair (L, A) where:*

(i) L is a Lie algebra (the closed colour, from the Lie *component);*
(ii) A is an associative algebra (the open colour, from the Ass *component);*
(iii) L acts on A by derivations, through the mixed sector;
(iv) the open-to-closed direction is zero.

The chiral Koszul dual $\mathcal{A}^!_{\mathrm{line}}$ of a chiral algebra carries exactly this structure: a Lie bracket (the Sklyanin bracket from the closed colour) and an associative product (the Yangian product from the open colour), with the Lie bracket acting as derivations.

Proof. This is a direct consequence of Theorem 4.1. The Lie cooperad dualises to the Lie operad on algebras; the Ass cooperad dualises to the Ass operad on algebras; the shuffle mixed sector dualises to the derivation condition. □

Remark 4.4 (The asymmetry principle)*.* The Koszul dual exchanges Com (one-dimensional, commutative) with Lie ($(n-1)!$-dimensional, non-commutative) on the closed colour, while preserving Ass on the open colour. This asymmetry is the operadic origin of the structural difference between the closed sector and the line side: the closed colour is governed by Com (factorisation on $\mathrm{Ran}(X)$ is symmetric), while the line-side Koszul dual is governed by Lie (the Sklyanin bracket, the Yangian Lie structure). The open colour, governed by Ass on both sides, is the shared associative structure: the E_1-algebra on the boundary equals the E_1-algebra on the line side.

5. PVA DESCENT: A_∞ OPERATIONS FROM $\mathrm{FM}_n(\mathbb{C})$

5.1. The A_∞ operations.

The bar differential $D_{\mathcal{A}}$ on $\overline{B}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ is a coderivation of the cofree coalgebra. By the universal property of cofree coalgebras, a coderivation is determined by its composition with the cogenerator projection $\mathrm{pr}\colon T^c(s^{-1}\bar{\mathcal{A}}) \to s^{-1}\bar{\mathcal{A}}$. The resulting sequence of multilinear maps

$$m_k\colon \mathcal{A}^{\otimes k} \longrightarrow \mathcal{A}((\lambda_1))\cdots((\lambda_{k-1})), \qquad k \geq 1, \quad |m_k| = 2-k,$$

are the A_∞ operations. Here $\lambda_j = z_j - z_k$ are spectral parameters recording the holomorphic separations of the insertion points on X.

The identity $D^2_{\mathcal{A}} = 0$ expands into the *Stasheff relations*: for each $n \geq 1$,

$$\sum_{\substack{i+j=n+1\\ i,j\geq 1}} \sum_{s=0}^{n-j} (-1)^{\epsilon(s,j)}\ m_i\big(a_1,\ldots,a_s,\ m_j(a_{s+1},\ldots,a_{s+j}),\ a_{s+j+1},\ldots,a_n\big) \ =\ 0, \tag{5.1}$$

where $\epsilon(s,j) = \sum_{t=1}^{s}(|a_t| - 1)$ is the Koszul sign from commuting m_j past the desuspended elements.

5.2. **Stasheff relations as Stokes' theorem.** Each Stasheff relation at degree n is Stokes' theorem on the Fulton–MacPherson space $\mathrm{FM}_n(\mathbb{C})$. The boundary $\partial\,\mathrm{FM}_n(\mathbb{C})$ is stratified by collision patterns: each codimension-one stratum D_S (indexed by a subset $S \subseteq \{1, \ldots, n\}$ with $|S| \geq 2$) records the collision of the points indexed by S. The contribution of D_S to the Stasheff relation is $m_{n-|S|+1}$ composed with $m_{|S|}$, where $m_{|S|}$ acts on the colliding points and $m_{n-|S|+1}$ acts on the result together with the remaining points.

Proposition 5.1 (Stasheff = Stokes). *The n-th Stasheff relation* (5.1) *is the identity*

$$\int_{\mathrm{FM}_n(\mathbb{C})} d\omega_n \;=\; \int_{\partial\,\mathrm{FM}_n(\mathbb{C})} \omega_n \;=\; \sum_{|S|\geq 2} \int_{D_S} \omega_n,$$

where ω_n is the chain-level integrand of m_n contracted with the weight forms $\eta_{ij} = d\log(z_i - z_j)$ and OPE coefficients. The identity $\partial^2 = 0$ on $\mathrm{FM}_n(\mathbb{C})$ (each codimension-two face is reached by exactly two codimension-one faces with opposite orientations) ensures $D^2_{\mathcal{A}} = 0$.

Proof. The bar differential is defined as a sum of boundary integrals on $\mathrm{FM}_n(\mathbb{C})$. The map m_n at degree n is the integral over the top-dimensional stratum of $\mathrm{FM}_n(\mathbb{C})$; $m_i \circ m_j$ contributions arise from boundary strata. Stokes' theorem identifies the sum of boundary contributions with the differential of the bulk integral. The identity $d^2 = 0$ on differential forms, applied to the FM compactification (where boundary strata compose correctly), gives $D^2_{\mathcal{A}} = 0$. □

We spell out the first four cases.

5.2.1. *$n = 1$: the BRST differential.* $\mathrm{FM}_1(\mathbb{C})$ is a single point with empty boundary. The Stasheff relation is $m_1 \circ m_1 = 0$: the map $m_1 = Q$ is a differential, $Q^2 = 0$. This is the BRST condition.

5.2.2. *$n = 2$: the Leibniz rule.* $\mathrm{FM}_2(\mathbb{C})$, modulo translations, is a point; its boundary is the single collision stratum $z_1 = z_2$. The Stasheff relation is

$$m_1(m_2(a,b)) \;+\; m_2(m_1(a),b) \;+\; (-1)^{|a|}\, m_2(a, m_1(b)) \;=\; 0:$$

the binary product m_2 is a chain map with respect to m_1.

5.2.3. *$n = 3$: homotopy associativity.* The Stasheff associahedron K_3 is a closed interval $[0,1]$, the quotient of $\mathrm{FM}_3(\mathbb{C})$ by the affine group. Its two endpoints are the collision strata $D_{12} = \{z_1 \to z_2\}$ and $D_{23} = \{z_2 \to z_3\}$. The Stasheff relation is:

$$m_2(m_2(a,b),c) - m_2(a, m_2(b,c)) \;=\; m_1(m_3(a,b,c)) + m_3(m_1(a),b,c) + (-1)^{|a|}\, m_3(a, m_1(b), c) + (-1)^{|a|+|b|}\, m_3(a,b, \tag{5.2}$$

The left-hand side is the associativity defect of m_2. The right-hand side is the boundary of the homotopy m_3: the binary product is associative up to the homotopy m_3, which is the integral over the interval K_3.

When $m_3 = 0$ (Heisenberg, free fermion), the relation reduces to strict associativity. When $m_3 \neq 0$ (affine Kac–Moody, Virasoro), the homotopy is the Jacobiator on K_3.

5.2.4. *$n = 4$: the Stasheff pentagon.* The associahedron K_4 is a compact polygon with five boundary faces, corresponding to the collision strata D_{12}, D_{23}, D_{34}, D_{123}, D_{234}. The Stasheff relation involves m_2, m_3, and m_4, with m_4 as the integral over the interior of K_4. For $\mathcal{W}_3$, the operation m_4 is non-zero, with the coefficient $32/(22 + 5c)$ entering through the $WW\Lambda$ channel (Section 7, third computation).

5.3. PVA descent on cohomology.

Definition 5.2 (Shifted Poisson vertex algebra shadow). A (-1)-*shifted Poisson vertex algebra shadow* over $\mathbb{k}$ is a commutative differential algebra $(V, \cdot, \partial)$ equipped with a λ-bracket $\{a_\lambda b\} \in V[\lambda]$ satisfying the following five axioms:

(PVA1) **Sesquilinearity.** $\{\partial a_\lambda b\} = -\lambda\{a_\lambda b\}$ and $\{a_\lambda \partial b\} = (\lambda + \partial)\{a_\lambda b\}$.
(PVA2) **Skew-symmetry.** $\{a_\lambda b\} = -\{b_{-\lambda-\partial} a\}$.
(PVA3) **Jacobi identity.** $\{a_\lambda\{b_\mu c\}\} - \{b_\mu\{a_\lambda c\}\} = \{\{a_\lambda b\}_{\lambda+\mu} c\}$.
(PVA4) **Left Leibniz rule.** $\{a_\lambda bc\} = \{a_\lambda b\}c + b\{a_\lambda c\}$.
(PVA5) **Commutativity.** $a \cdot b = b \cdot a$.

Theorem 5.3 (Shifted PVA descent). *Let $\mathcal{A}$ be a logarithmic $\mathsf{SC}^{\mathrm{ch,top}}$-algebra on a smooth curve X, with the regular and singular kernels separated, and let $H = H^\bullet(\mathcal{A}, m_1)$ be the cohomology with respect to the internal differential $m_1 = Q$. Suppose (H, p, ι, h) is a deformation retract of $(\mathcal{A}, m_1)$ onto H, with the side conditions $h^2 = h\iota = ph = 0$.*

Then:

(a) The homotopy transfer of the A_∞ structure $\{m_k\}$ to H produces transferred operations $\{m_k^H\}_{k\geq 1}$ satisfying the Stasheff relations.
(b) The transferred binary operation m_2^H decomposes as
$$m_2^H(a, b; \lambda) \;=\; a \cdot b + \{a_\lambda b\},$$
where $a \cdot b = m_2^H(a, b; 0)$ is a commutative associative product on H and $\{a_\lambda b\}$ is a λ-bracket.
(c) The pair $(H, \cdot, \{-_\lambda -\})$ is a (-1)-shifted Poisson vertex algebra shadow in the sense of Definition 5.2, with shifted Koszul signs.
(d) The higher transferred operations m_k^H for $k \geq 3$ encode the homotopy corrections to the chain-level shadow formality problem. Class G is the strict chain-level vanishing case, not the general cohomological PVA-descent statement.

Proof outline. **Part (a)** follows from the homotopy transfer theorem for A_∞ algebras (Kadeishvili [9], Merkulov [15]), applied through the homotopy-Koszulity of $\mathsf{SC}^{\mathrm{ch,top}}$ (Theorem 3.3).

Part (b). The binary operation m_2 has two sources: the constant term in λ (the normally-ordered product, which is commutative on cohomology) and the polynomial part in λ (the λ-bracket, which records the OPE singular part). On cohomology, the constant term survives as a commutative product because the Σ_n-equivariance of the chiral algebra (for E_∞-chiral algebras, all standard vertex algebras) forces $m_2^H(a, b; 0) = m_2^H(b, a; 0)$.

Part (c). The five PVA axioms are the cohomological shadows of the Stasheff relations:

- Sesquilinearity (PVA1) is the $n = 2$ Stasheff relation applied to ∂a and a, using the translation covariance of the chiral algebra.
- Skew-symmetry (PVA2) follows from the Σ_2-equivariance of $\mathrm{FM}_2(\mathbb{C})$ (the antipodal map $z \mapsto -z$ on the boundary circle).
- The Jacobi identity (PVA3) is the $n = 3$ Stasheff relation (5.2) on cohomology, where m_3 disappears (it is exact on H) and the associativity defect becomes the Jacobiator.
- The Leibniz rule (PVA4) is the compatibility of m_2 with the product structure, inherited from the factorisation property of the D-module.
- Commutativity (PVA5) is the cohomological consequence of E_∞-locality, as in part (b).

Part (d). If $\mathcal{A}$ is formal (all $m_k = 0$ for $k \geq 3$), then m_2 is strictly associative and the PVA structure on H is the complete algebraic datum. If $\mathcal{A}$ is non-formal (class L, C, or M), the higher operations m_k^H carry the residual A_∞ data that the PVA structure does not capture. □

5.4. **The five PVA axioms from geometry.** We record the geometric origin of each PVA axiom in detail.

Proposition 5.4 (Geometric origin of PVA axioms)**.** *The five PVA axioms correspond to geometric properties of the Fulton–MacPherson spaces as follows:*

Axiom	*FM property*	*Stasheff*
Sesquilinearity	*Translation covariance of* $\mathrm{FM}_2(\mathbb{C})$	$n = 2$
Skew-symmetry	Σ_2*-action on* $\mathrm{FM}_2(\mathbb{C})$	$n = 2$
Jacobi	*Arnold relation on* $\mathrm{FM}_3(\mathbb{C})$	$n = 3$
Leibniz	*Factorisation on* $\mathrm{FM}_2(\mathbb{C}) \times \mathrm{FM}_1(\mathbb{C})$	$n = 2$
Commutativity	E_∞*-locality (*Σ_n*-equivariance)*	*all* n

Proof. **Sesquilinearity.** The translation $z \mapsto z + c$ acts on $\mathrm{FM}_n(\mathbb{C})$ and preserves all boundary strata. On the bar complex, translation covariance is encoded by $D_{\mathcal{A}} \circ \partial = \partial \circ D_{\mathcal{A}}$, where ∂ is the translation operator on $\mathcal{A}$. Applied to a degree-two bar element $[s^{-1}\partial a \mid s^{-1}b]$, the Stasheff relation gives $m_2(\partial a, b; \lambda) = -\lambda\, m_2(a, b; \lambda)$ after extracting the λ-dependent part (the spectral parameter λ is conjugate to translation, so $\partial \leftrightarrow -\lambda$ under the Fourier-type correspondence between position and spectral variables). The right sesquilinearity follows by skew-symmetry.

Skew-symmetry. The involution $(z_1, z_2) \mapsto (z_2, z_1)$ on $\mathrm{FM}_2(\mathbb{C})$ acts on the boundary circle $E_{12} \cong S^1$ as the antipodal map. On the bar complex, this exchanges $[s^{-1}a \mid s^{-1}b]$ with $(-1)^{(|a|-1)(|b|-1)}[s^{-1}b \mid s^{-1}a]$. On the λ-bracket, the antipodal map sends λ to $-\lambda - \partial$ (the shift by ∂ arises from the chain rule in the Fourier transform), yielding $\{a_\lambda b\} = -\{b_{-\lambda-\partial} a\}$.

Jacobi identity. The Arnold relation on $\mathrm{FM}_3(\mathbb{C})$,

$$\eta_{12} \wedge \eta_{23} + \eta_{23} \wedge \eta_{31} + \eta_{31} \wedge \eta_{12} \;=\; 0,$$

where $\eta_{ij} = d\log(z_i - z_j)$, is the fundamental cohomological relation of the ordered configuration space. The $n = 3$ Stasheff relation on cohomology (where m_3 vanishes) reduces to

$$m_2(m_2(a, b), c) - m_2(a, m_2(b, c)) = 0 \quad (\text{modulo } m_3\text{-exact terms}).$$

Expanding m_2 into its product and λ-bracket components, the λ-bracket part of this relation is exactly the PVA Jacobi identity. The Arnold relation ensures the three cyclic terms cancel; this is the same mechanism that gives $D_{\mathcal{A}}^2 = 0$ at degree three.

Leibniz rule. The factorisation property of the chiral algebra gives a compatibility between the binary chiral product and the normally-ordered product. On $\mathrm{FM}_2(\mathbb{C}) \times \mathrm{FM}_1(\mathbb{C})$ (two colliding points and one spectator), the boundary stratum where the spectator approaches one of the colliding pair decomposes the interaction into two terms: $\{a_\lambda b\}c$ and $b\{a_\lambda c\}$. This is the Leibniz rule.

Commutativity. For E_∞-chiral algebras (all standard vertex algebras), the chiral product is Σ_n-equivariant: the factorisation D-module on $\mathrm{Ran}(X)$ is defined on *unordered* configurations. On cohomology, this forces the normally-ordered product to be commutative: $a \cdot b = b \cdot a$. For genuinely E_1-chiral algebras (Yangians, Etingof–Kazhdan quantum vertex algebras), commutativity fails and the PVA axiom must be replaced by a noncommutative variant. □

6. The resolvent principle

The A_∞ operations $\{m_k\}_{k\geq 2}$ are not independent. They are determined by a single datum: the binary product m_2.

6.1. The tree-level bootstrap equation.

Let (H, p, ι, h) be a deformation retract of $(\mathcal{A}, m_1)$ onto cohomology $H = H^\bullet(\mathcal{A}, m_1)$, with $h^2 = h\iota = ph = 0$.

Definition 6.1 (Dressed propagator)**.** The *dressed propagator* $\Phi\colon T^+(H) \to \mathcal{A}$ is defined by the fixed-point equation

$$\Phi \;=\; \iota + h \circ m_2 \circ \Phi^{\otimes 2}. \tag{6.1}$$

Here $T^+(H) = \bigoplus_{k\geq 1} H^{\otimes k}$ is the reduced tensor module, $\iota\colon H \hookrightarrow \mathcal{A}$ is the inclusion, and $h\colon \mathcal{A} \to \mathcal{A}$ is the contracting homotopy.

Equation (6.1) is the tree-level bootstrap: the minimal-model propagator from cohomology H into the chain algebra $\mathcal{A}$ is determined self-consistently from the free embedding ι and the interaction vertex m_2. Iteration produces the tree formula.

6.2. The tree formula.

Theorem 6.2 (Resolvent principle)**.** *The transferred A_∞ operations on H are given by*

$$m_k^H \;=\; \sum_{T\in \mathrm{PRT}_k} p \circ \mathfrak{m}(T) \circ \iota^{\otimes k},$$

where:

- *(i)* PRT_k *is the set of planar rooted binary trees with k leaves;*
- *(ii)* $|\mathrm{PRT}_k| = C_{k-1}$*, the $(k-1)$-st Catalan number ($C_0 = 1$, $C_1 = 1$, $C_2 = 2$, $C_3 = 5, \dots$);*
- *(iii)* $\mathfrak{m}(T)$ *assigns the binary product m_2 to each internal vertex and the contracting homotopy h to each internal edge;*
- *(iv)* $p\colon \mathcal{A} \to H$ *is the projection and* $\iota\colon H \to \mathcal{A}$ *is the inclusion.*

Proof. The iteration of (6.1) at order k collects all ways of building a k-to-1 map from $H^{\otimes k}$ to $\mathcal{A}$ using the vertex m_2 and the propagator h. Each such composition pattern is a planar rooted binary tree with k leaves (one leaf per input from H), internal vertices labelled by m_2, and internal edges labelled by h. The projection p extracts the cohomology class.

The number of planar rooted binary trees with k leaves is the Catalan number C_{k-1}: this follows from the recursive decomposition of a tree into its left and right subtrees, which gives the recurrence $C_n = \sum_{i=0}^{n-1} C_i C_{n-1-i}$ with $C_0 = 1$.

The Stasheff relations for $\{m_k^H\}$ are verified by checking that the boundary contributions of each tree (where two adjacent vertices merge) produce the correct composition terms. This is the combinatorial content of the associahedron: the faces of K_n correspond to the degenerate trees. □

One equation, one binary input, C_{k-1} trees.

Remark 6.3 (Physical interpretation)*.* The tree formula is the perturbative expansion of the Lagrangian self-intersection $\mathcal{L} \times^h_{\mathcal{M}} \mathcal{L}$ in a shifted symplectic stack. Each tree is a Feynman diagram with vertices m_2 and propagators h. The sum over PRT_k is the tree-level scattering amplitude at multiplicity k. Loop corrections (genus $g \geq 1$) arise from the curvature $d^2_{\mathrm{fib}} = \kappa(\mathcal{A}) \cdot \omega_g$, which obstructs the tree-level bootstrap at higher genus.

7. Three computations

7.1. First computation: Heisenberg (formal, class G). Let $\mathcal{A} = \mathcal{H}_k$ be the Heisenberg algebra at level k, with generator J and OPE

$$J(z)\,J(w) \;\sim\; \frac{k}{(z-w)^2}.$$

Proposition 7.1 (Heisenberg A_∞ structure)**.** *The A_∞ structure of $\mathcal{H}_k$ is:*

(i) $m_1 = 0$ *(no internal differential).*

(ii) $m_2(J,J;\lambda) = k\lambda$ *(the double-pole residue, absorbed by $d\log$: the OPE has a double pole, and the $d\log(z_1 - z_2)$ kernel reduces the pole order by one, producing a simple-pole collision residue k times the spectral parameter).*

(iii) $m_k = 0$ *for all $k \geq 3$ (no composites, no higher poles).*

The algebra is formal: the A_∞ structure is concentrated in m_2.

Proof. The augmentation-ideal projected binary product $\bar{\mu}_2(J,J) = 0$: the OPE $J(z)J(w) \sim k/(z-w)^2$ has singular part a scalar (the identity), which projects to zero in $\bar{\mathcal{A}} = \ker(\varepsilon)$. Therefore the bar differential on $\overline{B}(\mathcal{H}_k)$ vanishes identically: all bar elements are $D_{\mathcal{A}}$-closed.

The operation $m_2(J,J;\lambda) = k\lambda$ is obtained from the full OPE by the $d\log$ extraction: the coefficient of $1/(z_1-z_2)^2$ in the OPE is k, and the spectral parameter $\lambda = z_1 - z_2$ after Fourier transform gives $k\lambda$.

Since $\bar{\mu}_2 = 0$ on $\bar{\mathcal{A}}$, the degree-n bar differential involves only the identity-valued OPE residue, which lives outside $\bar{\mathcal{A}}$. All m_k for $k \geq 3$ vanish. □

Corollary 7.2 (Heisenberg PVA)**.** *On $H^\bullet(\mathcal{H}_k) = \Bbbk[J]$, the PVA structure is:*

(i) Product: $J^n \cdot J^m = J^{n+m}$ *(polynomial ring).*

(ii) λ-bracket: $\{J_\lambda J\} = k\lambda$.

(iii) All higher operations $m_k^H = 0$ *for* $k \geq 3$.

This is a PVA of rank one. The classical r-matrix is $r(z) = k/z$.

Verification: at $k = 0$, the r-matrix vanishes ($r(z) = 0$) and the algebra reduces to a commutative ring with trivial λ-bracket: the abelian limit.

The modular characteristic is $\kappa(\mathcal{H}_k) = k$: the same scalar governs the OPE, the λ-bracket, the r-matrix, and the genus-one curvature $d^2_{\mathrm{fib}} = k \cdot \omega_1$.

Remark 7.3 (Shadow depth zero). The Heisenberg algebra belongs to class G in the shadow depth classification: $d_{\mathrm{alg}} = 0$. The shadow tower terminates at $\kappa = k$; the discriminant $\Delta = 8\kappa S_4 = 0$ (since $S_4 = 0$ for the Heisenberg) confirms the finite tower. The SC-formality condition is satisfied: $\mathcal{H}_k$ is SC-formal because $m_k = 0$ for $k \geq 3$.

7.2. Second computation: Kac–Moody at level k (Lie-transverse, class L). Let $\mathcal{A} = V_k(\mathfrak{g})$ be the vacuum module of the affine Kac–Moody algebra at level $k \in \mathbb{C} \setminus \{-h^\vee\}$, with generators $\{J^a\}$ ($a = 1, \dots, d = \dim\mathfrak{g}$) and OPE

$$J^a(z)\,J^b(w) \;\sim\; \frac{k\,\kappa^{ab}}{(z-w)^2} \;+\; \frac{f^{ab}{}_c\,J^c(w)}{z-w}, \tag{7.1}$$

where κ^{ab} is the normalised Killing form and $f^{ab}{}_c$ are the structure constants.

Proposition 7.4 (Kac–Moody A_∞ structure)**.** *The A_∞ structure of $V_k(\mathfrak{g})$ is:*

(i) $m_1 = 0$.

(ii) $m_2(J^a, J^b; \lambda) = f^{ab}{}_c\, J^c + k\, \kappa^{ab}\, \lambda$. *The simple-pole coefficient* $f^{ab}{}_c J^c$ *is the Lie bracket; the double-pole coefficient* $k\,\kappa^{ab}$ *contributes the* λ*-dependent term.*

(iii) $m_3 \neq 0$*: the Jacobiator homotopy on* $\mathrm{FM}_3(\mathbb{C})$*. The three codimension-one boundary strata of* $\mathrm{FM}_3(\mathbb{C})$ *contribute* $m_2(m_2(-,-),-)$ *terms; their failure to cancel (the Jacobi identity defect from the simple-pole bracket) is corrected by* m_3*.*

(iv) $m_k = 0$ *for all* $k \geq 4$*, by dimension counting on* $\mathrm{FM}_k(\mathbb{C})$*: the OPE has at most a double pole, so the chain-level integrands vanish for* $k \geq 4$*.*

Proof. The binary operation extracts from the OPE (7.1): the residue at $z_1 = z_2$ with the $d\log(z_1 - z_2)$ kernel gives the simple-pole coefficient $f^{ab}{}_c J^c$ (the Lie bracket), and the double-pole coefficient $k\kappa^{ab}$ contributes the λ-linear term.

The existence of $m_3 \neq 0$ follows from the failure of strict associativity: $m_2(m_2(J^a, J^b), J^c) - m_2(J^a, m_2(J^b, J^c)) = f^{ab}{}_e f^{ec}{}_d J^d - f^{bc}{}_e f^{ae}{}_d J^d \neq 0$ (the Jacobi defect). The homotopy m_3 is the integral over $K_3 = [0,1]$ that interpolates between the two boundary contributions.

The vanishing $m_k = 0$ for $k \geq 4$ follows from the pole structure: the OPE has at most double poles, and the weight forms η_{ij} absorb one pole order each. For $k \geq 4$ insertion points, the integrand on $\mathrm{FM}_k(\mathbb{C})$ involves at most $(k-1)$ weight forms but the pole excess is at most 2, which is consumed at $k = 3$. For $k \geq 4$, the integrand has no residual singularity and the FM integral vanishes. □

Corollary 7.5 (Kac–Moody PVA). *On* $H^\bullet(V_k(\mathfrak{g})) = \Bbbk[J^a : a = 1, \ldots, d]$*, the PVA structure is:*

(i) Product: polynomial ring in d *generators.*

(ii) λ*-bracket:* $\{J^a_\lambda J^b\} = f^{ab}{}_c\, J^c + k\, \kappa^{ab}\, \lambda$*.*

(iii) Classical r-matrix: $r(z) = k\, \Omega_{\mathrm{tr}}/z$ *in the trace-form convention, where* $\Omega_{\mathrm{tr}} = \sum_a J^a \otimes J_a$ *is the split Casimir.*

Verification: at $k = 0$*, the r-matrix vanishes (*$r(z) = 0$*); the algebra reduces to the commutative ring with Lie bracket* $\{J^a_\lambda J^b\} = f^{ab}{}_c J^c$*, which is still a well-defined PVA (the abelian limit of the level).*

In the KZ convention: $r(z) = \Omega/((k + h^\vee)z)$*. Bridge:* $k\, \Omega_{\mathrm{tr}} = \Omega/(k + h^\vee)$ *at generic k.*

The modular characteristic is

$$\kappa(V_k(\mathfrak{g})) \;=\; \frac{\dim(\mathfrak{g})\,(k + h^\vee)}{2h^\vee}.$$

At $k = 0$*:* $\kappa = \dim(\mathfrak{g})/2$ *(not zero: the pure vacuum anomaly). At* $k = -h^\vee$*:* $\kappa = 0$ *(critical level, the bar complex is flat).*

Remark 7.6 (Shadow depth one, class L). The affine Kac–Moody algebra belongs to class L: $d_{\mathrm{alg}} = 1$. The shadow tower extends to the cubic invariant $\mathfrak{C}(\hat{\mathfrak{g}}_k)$ (from the Lie structure constants) but terminates at shadow depth $r_{\max} = 3$. The quartic shadow invariant $S_4 = 0$, so the discriminant $\Delta = 8\kappa S_4 = 0$: the shadow tower is finite (class L terminates at depth 3). The algebra is *not* SC-formal: $m_3 \neq 0$.

7.3. **Third computation: $\mathcal{W}_3$ (infinite tower, class M).** The $\mathcal{W}_3$ algebra has two generators: the stress tensor T of conformal weight 2 and a spin-3 current W of conformal weight 3.

Proposition 7.7 ($\mathcal{W}_3$ A_∞ structure). *The* A_∞ *structure of* $\mathcal{W}_3$ *satisfies:*

(i) $m_1 = 0$.

(ii) $m_2(T, T; \lambda) = (c/12)\lambda^3 + 2T\lambda + \partial T$ *(the full* TT *OPE in divided-power parametrisation; the coefficient* $c/12$ *matches* $T_{(3)}T = c/2$ *via the divided power* $\lambda^{(3)} = \lambda^3/3!$*).*

(iii) $m_2(T, W; \lambda) = 3W\lambda + \partial W$.

(iv) The WW operation involves the composite field $\Lambda = {:}(TT){:} - \frac{3}{10}\partial^2 T$ at the λ-linear term:

$$m_2(W, W; \lambda) \;=\; \frac{c}{360}\lambda^5 + \frac{1}{3}T\lambda^3 + \frac{1}{2}(\partial T)\lambda^2 + \Big(\frac{3}{10}\partial^2 T + \frac{32}{22+5c}\,\Lambda\Big)\lambda + \cdots$$

The coefficient $32/(22+5c)$ diverges at $c = -22/5$ (the Lee–Yang value).

(v) $m_3 \neq 0$: the Jacobiator homotopy, involving Λ.

(vi) $m_4 \neq 0$: the quartic operation from the $\mathrm{FM}_4(\mathbb{C})$ boundary decomposition, with $32/(22+5c)$ entering through the $WW\Lambda$ channel.

(vii) The A_∞ structure does not terminate: $m_k \neq 0$ for all $k \geq 3$.

Proof sketch. The binary operations are extracted from the Zamolodchikov $\mathcal{W}_3$ OPE [17] using the $d\log$ kernel and divided-power parametrisation. The non-vanishing of m_3 follows from the non-Lie structure of the $\mathcal{W}_3$ bracket (the bracket involves the composite field Λ, which is not a generator). The non-vanishing of m_4 follows from the Stasheff pentagon: the m_3-m_3 composition on ∂K_4 does not cancel (the Λ channel provides a non-trivial residue on the interior of K_4). The infinite tower $m_k \neq 0$ for all k follows by induction: each m_k generates a non-vanishing correction at degree $k+1$ through the Stasheff cascade, driven by the non-primitive composite Λ. □

Corollary 7.8 ($\mathcal{W}_3$ PVA). *On cohomology, the PVA structure is the Zamolodchikov PVA:*

$$\begin{aligned}
\{T_\lambda T\} &= \frac{c}{12}\lambda^3 + 2T\lambda + \partial T,\\
\{T_\lambda W\} &= 3W\lambda + \partial W,\\
\{W_\lambda W\} &= \frac{c}{360}\lambda^5 + \frac{1}{3}T\lambda^3 + \frac{1}{2}(\partial T)\lambda^2 + \Big(\frac{3}{10}\partial^2 T + \frac{32}{22+5c}\,\Lambda\Big)\lambda + \cdots
\end{aligned}$$

The modular characteristic is $\kappa(\mathcal{W}_3) = 5c/6$: from the formula $\kappa(\mathcal{W}_N) = c\,(H_N - 1)$ with $H_N = \sum_{j=1}^{N} 1/j$, at $N = 3$ we have $H_3 = 1 + 1/2 + 1/3 = 11/6$, so $\kappa = c \cdot 5/6$.

Verification at $N = 2$: $H_2 = 3/2$, $H_2 - 1 = 1/2$, so $\kappa(\mathcal{W}_2) = c/2$, which matches the Virasoro formula $\kappa(\mathrm{Vir}_c) = c/2$ (since $\mathcal{W}_2 = \mathrm{Vir}$).

Remark 7.9 (Class M: infinite shadow depth). The $\mathcal{W}_3$ algebra belongs to class M: $d_{\mathrm{alg}} = \infty$. The shadow tower $S_2, S_3, S_4, \ldots$ is infinite, driven by the non-primitive composite Λ. The generating depth $d_{\mathrm{gen}} = 3$ (the operation m_3 generates all higher m_k via the Stasheff cascade), but the algebraic depth is infinite (all $m_k \neq 0$). The discriminant $\Delta = 8\kappa S_4 \neq 0$ for $c \neq 0$ and $c \neq -22/5$, confirming the infinite tower.

This infinite depth is the algebraic shadow of the fact that three-dimensional gravity with $\mathcal{W}_3$ symmetry requires an infinite number of graviton vertices: the A_∞ structure is the perturbative expansion of the bulk theory.

Algebra	**Class**	d_{alg}	κ	$m_{k\geq 3}$	**PVA type**
$\mathcal{H}_k$	G	0	k	$= 0$	free rank-1
$V_k(\mathfrak{g})$	L	1	$\frac{d(k+h^\vee)}{2h^\vee}$	$m_3 \neq 0, m_{\geq 4} = 0$	affine, Lie-transverse
$\mathcal{W}_3$	M	∞	$5c/6$	all $\neq 0$	Zamolodchikov, infinite

8. Heptagon, topologization scope, and PVA descent

The operad $\mathsf{SC}^{\mathrm{ch,top}}$ is controlled by three tests: the Vol II $\mathsf{SC}^{\mathrm{ch,top}}$ heptagon (seven edges routing bar, cobar, Drinfeld centre, derived centre, Swiss-cheese, PTVV, and celestial holography), the four-class topologisation scope tree, and the Khan–Zeng PVA descent for freely-generated Poisson vertex algebras.

Theorem 8.1 (The $\mathsf{SC}^{\mathrm{ch,top}}$ heptagon)**.** *The two-coloured operad* $\mathsf{SC}^{\mathrm{ch,top}}$ *sits at the hub of a heptagon of seven equivalent presentations, each edge a named theorem:*

(1) Operadic hub: $\mathsf{SC}^{\mathrm{ch,top}}$ *itself (this paper).*
(2) Koszul dual: $(\mathsf{SC}^{\mathrm{ch,top}})^! = (\mathsf{Lie}^c, \mathsf{Ass}^c, \text{shuffle-mixed})$ *(this paper, Theorem 4.1;* not *Koszul self-dual).*
(3) Chiral factorization: *the Beilinson–Drinfeld factorization algebra on* $\mathrm{Ran}(X)$ *with open* $\mathbb{R}$*-decoration.*
(4) BV–BRST: *the Costello–Gwilliam bulk–boundary BV complex.*
(5) Convolution: *the* L_∞*-algebra* $\mathrm{Hom}_\alpha(C, A)$ *with open-closed decoration.*
(6) Drinfeld centre: $Z(\mathrm{Rep}_{\mathrm{fact}}(\mathcal{A})) \simeq \mathrm{Rep}_{\mathrm{fact}}(Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}))^{E_2}$ *via categorified bar–cobar with half-braiding.*
(7) PTVV derived-AG: *the 1-shifted symplectic mapping stack* $\mathrm{Map}(X \times \mathbb{R}_{\geq 0}, B\,\mathsf{SC}\text{-Alg})$.

Each of the seven is a distinct theorem; the Vol II heptagon chapter (`sc_chtop_heptagon.tex`*) proves the seven edges as named theorems.*

Theorem 8.2 (Sugawara topologization by shadow class)**.** *The passage* $\mathsf{SC}^{\mathrm{ch,top}} \to E_3^{\mathrm{top}}$ *via Sugawara topologisation branches by shadow class:*

- *Class G (Heisenberg):* E_3^{top} *immediate; commutative input, Dunn on* F^{H_k}.
- *Class L (*$V_k(\mathfrak{g})$*,* $k \neq -h^\vee$*):* E_3^{top} proved on the original complex*, via Costello–Gwilliam factorization on* $\mathbb{R} \times \mathbb{C}$ *and explicit BRST primitives* $\eta_1^{(i)}, \eta_1^{(ii)}$ *giving* $[Q, \widetilde{G}_1] = T_{\mathrm{Sug}}$ *at strict chain level.*
- *Class C (*$\beta\gamma$*,* bc*):* E_3^{top} *via Friedan–Martinec–Shenker bosonisation reducing to class G.*
- *Class M (*Vir*,* $\mathcal{W}_N$*):* E_3^{top} weight-completed*, via half-BRST plus MC5 pattern; original-complex chain-level is the sole remaining open direction.*
- *Critical level* $k = -h^\vee$*:* E_2^{top} *(a dimension drop, not a failure; Sugawara is undefined at critical level).*

Without a conformal vector (or at critical level), the theory is stuck at $\mathsf{SC}^{\mathrm{ch,top}}$*; this is the generic situation (Heisenberg, lattice,* E_1*-chiral Yangian input, free fields without stress tensor), for which* $\mathsf{SC}^{\mathrm{ch,top}}$ *is the final answer and* E_3 *the special case.*

Theorem 8.3 (Khan–Zeng PVA descent on freely generated PVAs)**.** *Let* A *be a freely-generated Poisson vertex algebra with conformal vector. Then the* A_∞ *operations* $\{m_k\}$ *extracted by cogenerator projection from the bar differential of the associated graded* $\mathrm{gr}_{Li}(A)$ *descend to the five PVA axioms (sesquilinearity, skew-symmetry, Jacobi, Leibniz, commutativity) on cohomology. The descent covers the 3d Poisson sigma-model class of Khan–Zeng, comprising all freely-generated PVAs with conformal vector; in particular, it recovers the Vol II orbifold route via* $\mathbb{Z}/n$*-invariants preserving* E_n*-structure.*

Remark 8.4 (The bar complex is not an $\mathsf{SC}^{\mathrm{ch,top}}$-coalgebra)*.* As stated in Warning 1.1, the bar complex $\overline{B}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ is an E_1 chiral coassociative coalgebra, *not* an $\mathsf{SC}^{\mathrm{ch,top}}$-coalgebra. The $\mathsf{SC}^{\mathrm{ch,top}}$ structure lives on the derived centre pair $(C^\bullet_{\mathrm{ch}}(\mathcal{A}, \mathcal{A}), \mathcal{A})$; the bar complex serves as a resolution used to *compute* $C^\bullet_{\mathrm{ch}}(\mathcal{A}, \mathcal{A})$, not as an object carrying $\mathsf{SC}^{\mathrm{ch,top}}$-structure. The obstruction is formal: the bar differential is not a closed colour, deconcatenation is not an open colour, and two structures on one coalgebra do not produce the two colours of $\mathsf{SC}^{\mathrm{ch,top}}$. The $\mathsf{SC}^{\mathrm{ch,top}}$ structure emerges in $\mathrm{Hom}(B(\mathcal{A}), \mathcal{A})$, not in $B(\mathcal{A})$.

Remark 8.5 ($\mathsf{SC}^{\mathrm{ch,top}}$ is Koszul, not self-dual)*.* $(\mathsf{SC}^{\mathrm{ch,top}})^! \neq \mathsf{SC}^{\mathrm{ch,top}}$: closed-colour dimensions differ at $n \geq 3$ ($\dim \mathsf{Com}(n) = 1$ vs $\dim \mathsf{Lie}(n) = (n-1)!$). The duality *functor* is an involution; the operad is *not* self-dual. Livernet's proof gives Koszulity, not self-duality.

Remark 8.6 (Status of the comparison theorems). Theorem 8.1: the seven heptagon edges are *proved* in the Vol II heptagon chapter; the operadic hub (edges 1–3) is proved directly. Theorem 8.2: class G, L, C are unconditional on the original complex; class M is weight-completed. Theorem 8.3: unconditional for freely-generated PVAs; the general class M case reduces to the weight-completed topologization.

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